

6.5 Solution: (a) Using the expression for the momentum of the EM field and the Maxwell equations, we have

$$\mathbf{P}_{\text{field}} = \frac{1}{c^2} \int (\mathbf{E} \times \mathbf{H}) d^3x \quad (1)$$

$$\begin{aligned} &= -\frac{1}{c^2} \int (\nabla \Phi \times \mathbf{H}) d^3x \\ &= -\frac{1}{c^2} \int \left[\nabla \times (\Phi \mathbf{H}) - \Phi (\nabla \times \mathbf{H}) \right] d^3x \\ &= \frac{1}{c^2} \int \Phi \mathbf{J} d^3x - \frac{1}{c^2} \int \nabla \times (\Phi \mathbf{H}) d^3x \\ &= \frac{1}{c^2} \int \Phi \mathbf{J} d^3x - \frac{1}{c^2} \oint \mathbf{n} \times (\Phi \mathbf{H}) da, \end{aligned} \quad (2)$$

where we have used $\mathbf{E} = -\nabla \Phi$, $\nabla \times \mathbf{H} = \mathbf{J}$, $\nabla \times (\psi \mathbf{a}) = \nabla \psi \times \mathbf{a} + \psi (\nabla \times \mathbf{a})$, and the Stokes theorem to transform the volume integral to the surface integral in the last step. The surface integral can be dropped, as long as the integrand $\Phi \mathbf{H}$ falls faster than r^{-2} , which is the case for electric and magnetic fields generated from localized charge and current, as both of them fall as r^{-2} . Therefore, we can write the EM field momentum as

$$\mathbf{P}_{\text{field}} = \frac{1}{c^2} \int \Phi \mathbf{J} d^3x. \quad (3)$$

(b) We can apply the result from part (a), and notice that near the current distribution,

$$\Phi(\mathbf{x}) = \Phi(\mathbf{0}) + \nabla \Phi(\mathbf{0}) \cdot \mathbf{x} = \Phi(\mathbf{0}) - \mathbf{E}(\mathbf{0}) \cdot \mathbf{x},$$

the EM field momentum becomes

$$\mathbf{P}_{\text{field}} = \frac{\Phi(\mathbf{0})}{c^2} \int \mathbf{J}(\mathbf{x}) d^3x - \frac{1}{c^2} \mathbf{E}(\mathbf{0}) \cdot \int \mathbf{x} \mathbf{J}(\mathbf{x}) d^3x.$$

From section 5.6, we know that

$$\int \mathbf{J}(\mathbf{x}) d^3x = 0, \quad \mathbf{n} \cdot \int \mathbf{x} \mathbf{J}(\mathbf{x}) d^3x = -\frac{1}{2} \mathbf{n} \times \int [\mathbf{x} \times \mathbf{J}(\mathbf{x})] d^3x = -\mathbf{n} \times \mathbf{m}.$$

Then,

$$\mathbf{P}_{\text{field}} = -\frac{1}{c^2} E(\mathbf{0}) \mathbf{n} \cdot \int \mathbf{x} \mathbf{J}(\mathbf{x}) d^3x = \frac{1}{c^2} E(\mathbf{0}) \mathbf{n} \times \mathbf{m} = \frac{1}{c^2} E(\mathbf{0}) \times \mathbf{m}.$$

(c) For a uniform electric field, the surface integral term in Eq. (2) cannot be dropped, as $\Phi \sim r$ and $|\mathbf{H}| \sim r^{-2}$ by a crude estimation.

We can calculate the EM field momentum in two ways. First, using Eq. (1) and notice that the electric field is a uniform one, we can express the result as

$$\mathbf{P}_{\text{field}} = \frac{1}{c^2} \mathbf{E}_0 \times \int \mathbf{H} d^3x.$$

From Section 5.6, by Eq. (5.62), we also know

$$\int \mathbf{H} d^3x = \frac{2}{3} \mathbf{m}.$$

Then,

$$\mathbf{P}_{\text{field}} = \frac{2}{3c^2} \mathbf{E}_0 \times \mathbf{m}. \quad (4)$$

We can also determine the EM field momentum by explicit calculation of the surface integral. In Eq. (2), since $\Phi(\mathbf{x}) = -\mathbf{E}_0 \cdot \mathbf{x}$, the first term becomes

$$\frac{1}{c^2} \int \Phi \mathbf{J} d^3x = -\frac{1}{c^2} \mathbf{E}_0 \cdot \int \mathbf{x} \mathbf{J}(\mathbf{x}) d^3x,$$

which, from the result of part (b), is

$$\frac{1}{c^2} \int \Phi \mathbf{J} d^3x = \frac{1}{c^2} \mathbf{E}_0 \times \mathbf{m}.$$

For the second term, the surface integral, we can resort to the magnetic dipole moment approximation and express the magnetic field as

$$\mathbf{H} = \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{m}) - \mathbf{m}}{4\pi r^3},$$

and

$$\mathbf{n} \times (\Phi \mathbf{H}) = -\frac{\Phi(\mathbf{n} \times \mathbf{m})}{4\pi r^3}.$$

Now, the surface integral becomes

$$\oint \mathbf{n} \times (\Phi \mathbf{H}) da = \frac{1}{4\pi} \mathbf{E}_0 \cdot \int \mathbf{n}(\mathbf{n} \times \mathbf{m}) d\Omega = \frac{1}{4\pi} E_{0,i} m_c \epsilon_{abc} \hat{e}_a \int n_i n_b d\Omega,$$

where we have used the fact that $\mathbf{n} \times \mathbf{m} = \epsilon_{abc} \hat{e}_a n_b m_c$, with Einstein summation convention assumed. The integral with respect to the angles leads to

$$\int n_i n_b d\Omega = \frac{4\pi}{3} \delta_{ib},$$

and the surface integral becomes

$$\frac{1}{4\pi} \epsilon_{abc} \hat{e}_a E_{0,i} m_c \frac{4\pi}{3} \delta_{ib} = \frac{1}{3} \epsilon_{abc} \hat{e}_a E_{0,b} m_c = \frac{1}{3} \mathbf{E}_0 \times \mathbf{m}.$$

Finally, the EM field momentum is

$$\mathbf{P}_{\text{field}} = \frac{1}{c^2} \mathbf{E}_0 \times \mathbf{m} - \frac{1}{3c^2} \mathbf{E}_0 \times \mathbf{m} = \frac{2}{3c^2} \mathbf{E}_0 \times \mathbf{m}.$$

In the above, we have used magnetic dipole moment approximation to get the result. However, this is not satisfactory, as the EM field momentum (4) is an exact result and should not rely on any approximation. To this end, consider the magnetic field from a localized current distribution,

$$\mathbf{H} = \frac{1}{4\pi} \nabla \times \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' = \frac{1}{4\pi} \int \nabla \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) \times \mathbf{J}(\mathbf{x}') d^3x'.$$

Next, we can use the expansion

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \theta),$$

to express the gradient as, assuming $r_{<} = r'$ and $r_{>} = r$,

$$\nabla \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) = \sum_{l=0}^{\infty} \left(-(l+1) \frac{(r')^l}{r^{l+2}} P_l(\cos \theta) \hat{r} - \frac{(r')^l}{r^{l+2}} \sin \theta P'_l(\cos \theta) \hat{\theta} \right).$$